



Assignments

2D Arrays – 2



Q1. Write a program to print the elements of both the diagonals in a square matrix.

Input 1:

1 2 3

4 5 6

7 8 9

Output 1:

1 3

5

7 9

Q2. Write a program to rotate the matrix by 90 degrees anti-clockwise.

Input 1:

1 2 3

4 5 6

7 8 9

Output 1:

3 6 9

2 5 8

1 4 7

Q3. Write a program to print the matrix in wave form.

Input :

1 2 3

4 5 6

7 8 9

Output : 7 4 1 2 5 8 9 6 3

Q4. Given a positive integer n , generate a $n \times n$ matrix filled with elements from 1 to n^2 in spiral order.

Input 1: $n = 3$

Output 1: $[[1,2,3],[8,9,4],[7,6,5]]$

Input 2: $n = 1$

Output 2: $[[1]]$

Q5. Predict the output :

```
int main(){  
int a[][2] = {{1,2},{3,4}};  
int i, j;  
for (i = 0; i < 2; i++)  
for (j = 0; j < 2; j++)  
cout << a[i][j];  
return 0;  
}
```



SKILLS

Note:- Please try to invest time doing the assignments which are necessary to build a strong foundation. Do not directly Copy Paste using Google or ChatGPT. Please use your brain.